In [1]:
```
# Run this ONCE to install all libraries
!pip install pandas numpy matplotlib seaborn scikit-learn xgboost shap imbalanced-l
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pandas in c:\programdata\anaconda3\lib\site-packages
(2.3.3)
Requirement already satisfied: numpy in c:\programdata\anaconda3\lib\site-packages
(2.3.5)
Requirement already satisfied: matplotlib in c:\programdata\anaconda3\lib\site-packa
ges (3.10.6)
Requirement already satisfied: seaborn in c:\programdata\anaconda3\lib\site-packages
(0.13.2)
Requirement already satisfied: scikit-learn in c:\programdata\anaconda3\lib\site-pac
kages (1.7.2)
Requirement already satisfied: xgboost in c:\users\user\appdata\roaming\python\pytho
n313\site-packages (3.2.0)
Requirement already satisfied: shap in c:\users\user\appdata\roaming\python\python31
3\site-packages (0.50.0)
Requirement already satisfied: imbalanced-learn in c:\programdata\anaconda3\lib\site
-packages (0.14.0)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\programdata\anaconda3\li
b\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\programdata\anaconda3\lib\site-pac
kages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\programdata\anaconda3\lib\site-p
ackages (from pandas) (2025.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\programdata\anaconda3\lib\site
-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\programdata\anaconda3\lib\site-pac
kages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\programdata\anaconda3\lib\sit
e-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\programdata\anaconda3\lib\sit
e-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\programdata\anaconda3\lib\site-
packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\programdata\anaconda3\lib\site-packag
es (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\programdata\anaconda3\lib\site
-packages (from matplotlib) (3.2.5)
Requirement already satisfied: scipy>=1.8.0 in c:\programdata\anaconda3\lib\site-pac
kages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in c:\programdata\anaconda3\lib\site-pa
ckages (from scikit-learn) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\programdata\anaconda3\lib
\site-packages (from scikit-learn) (3.5.0)
Requirement already satisfied: tqdm>=4.27.0 in c:\programdata\anaconda3\lib\site-pac
kages (from shap) (4.67.1)
Requirement already satisfied: slicer==0.0.8 in c:\users\user\appdata\roaming\python
\python313\site-packages (from shap) (0.0.8)
Requirement already satisfied: numba>=0.54 in c:\programdata\anaconda3\lib\site-pack
ages (from shap) (0.62.1)
Requirement already satisfied: cloudpickle in c:\programdata\anaconda3\lib\site-pack
ages (from shap) (3.1.1)
Requirement already satisfied: typing-extensions in c:\programdata\anaconda3\lib\sit
e-packages (from shap) (4.15.0)
Requirement already satisfied: llvmlite<0.46,>=0.45.0dev0 in c:\programdata\anaconda
3\lib\site-packages (from numba>=0.54->shap) (0.45.1)
Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\site-package
```

```
s (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: colorama in c:\programdata\anaconda3\lib\site-package
s (from tqdm>=4.27.0->shap) (0.4.6)
```

In [2]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix, roc_auc_score
from imblearn.over_sampling import SMOTE
import shap
from xgboost import XGBClassifier
import warnings
warnings.filterwarnings('ignore')

print("🎉 Phase 1 Complete! All libraries ready!")
```

```
🎉 Phase 1 Complete! All libraries ready!
```

In [3]:
```python
# ── PHASE 2: Load Dataset ──────────────────────────────
df = pd.read_csv(r'C:\ML Project\WA_Fn-UseC_-Telco-Customer-Churn.csv')

print(' Dataset loaded successfully!')
print(f'   Rows    : {df.shape[0]}')
print(f'   Columns : {df.shape[1]}')
```

```
Dataset loaded successfully!
  Rows    : 7043
  Columns : 21
```

In [4]:
```python
# First 5 rows of the dataset
df.head()
```

Out[4]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	Multipl
0	7590-VHVEG	Female	0	Yes	No	1	No	No
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	No
4	9237-HQITU	Female	0	No	No	2	Yes	

5 rows × 21 columns

In [5]:

```
# Data types and column information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   customerID        7043 non-null   object 
 1   gender            7043 non-null   object 
 2   SeniorCitizen     7043 non-null   int64  
 3   Partner           7043 non-null   object 
 4   Dependents        7043 non-null   object 
 5   tenure            7043 non-null   int64  
 6   PhoneService      7043 non-null   object 
 7   MultipleLines     7043 non-null   object 
 8   InternetService   7043 non-null   object 
 9   OnlineSecurity    7043 non-null   object 
 10  OnlineBackup      7043 non-null   object 
 11  DeviceProtection  7043 non-null   object 
 12  TechSupport       7043 non-null   object 
 13  StreamingTV       7043 non-null   object 
 14  StreamingMovies   7043 non-null   object 
 15  Contract          7043 non-null   object 
 16  PaperlessBilling  7043 non-null   object 
 17  PaymentMethod     7043 non-null   object 
 18  MonthlyCharges    7043 non-null   float64 
 19  TotalCharges      7043 non-null   object 
 20  Churn             7043 non-null   object 
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

In [6]:
```
# Statistical summary of numeric columns
df.describe().round(2)
```

Out[6]:

	SeniorCitizen	tenure	MonthlyCharges
count	7043.00	7043.00	7043.00
mean	0.16	32.37	64.76
std	0.37	24.56	30.09
min	0.00	0.00	18.25
25%	0.00	9.00	35.50
50%	0.00	29.00	70.35
75%	0.00	55.00	89.85
max	1.00	72.00	118.75

In [7]:
```
# Check missing values in each column
df.isnull().sum()
```

Out[7]:
```
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

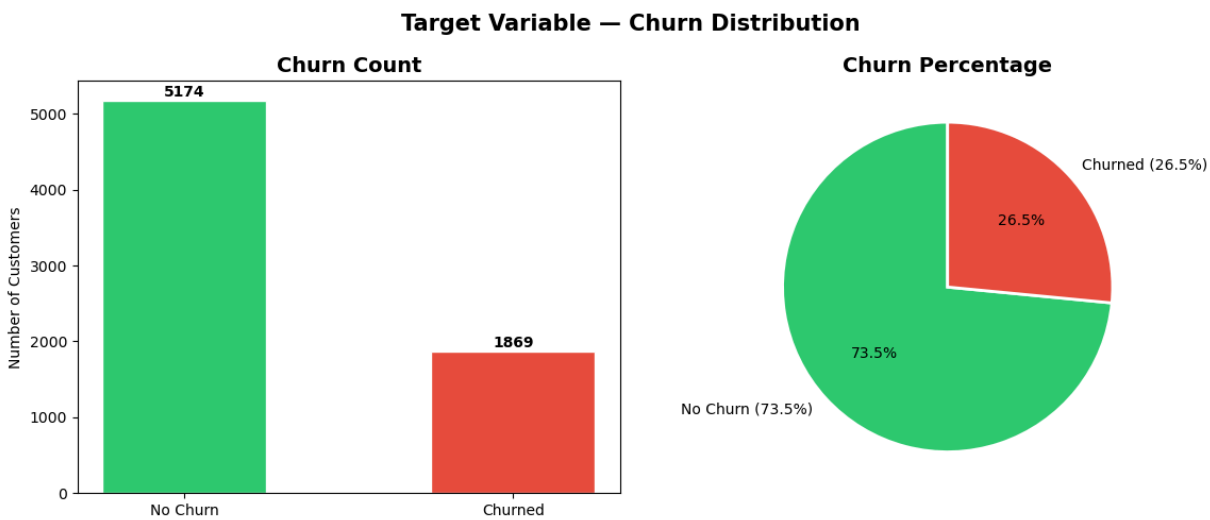
In [8]:
```
print(f'Duplicate Rows: {df.duplicated().sum()}')
```

```
Duplicate Rows: 0
```

In [9]:
```
print(df['Churn'].value_counts())
print()
print(df['Churn'].value_counts(normalize=True).mul(100).round(1).astype(str) + '%')
```

```
Churn
No     5174
Yes    1869
Name: count, dtype: int64

Churn
No     73.5%
Yes    26.5%
Name: proportion, dtype: object
```

In [10]:
```
# ── PHASE 3 | PART A — Univariate Analysis ──────────────
# Chart 1: Churn Distribution

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# Bar Chart
colors = ['#2ecc71', '#e74c3c']
axes[0].bar(['No Churn', 'Churned'], [5174, 1869], color=colors, width=0.5, edgecol
axes[0].set_title('Churn Count', fontsize=14, fontweight='bold')
axes[0].set_ylabel('Number of Customers')
axes[0].text(0, 5174 + 50, '5174', ha='center', fontweight='bold')
axes[0].text(1, 1869 + 50, '1869', ha='center', fontweight='bold')

# Pie Chart
axes[1].pie([5174, 1869], labels=['No Churn (73.5%)', 'Churned (26.5%)'],
            colors=colors, autopct='%1.1f%%', startangle=90,
            wedgeprops={'edgecolor': 'white', 'linewidth': 2})
axes[1].set_title('Churn Percentage', fontsize=14, fontweight='bold')

plt.suptitle('Target Variable — Churn Distribution', fontsize=15, fontweight='bold'
plt.tight_layout()
plt.show()

print(' OBSERVATION: 26.5% customers churned vs 73.5% stayed.')
print('   Class imbalance exists — SMOTE will be applied in Phase 4.')
```

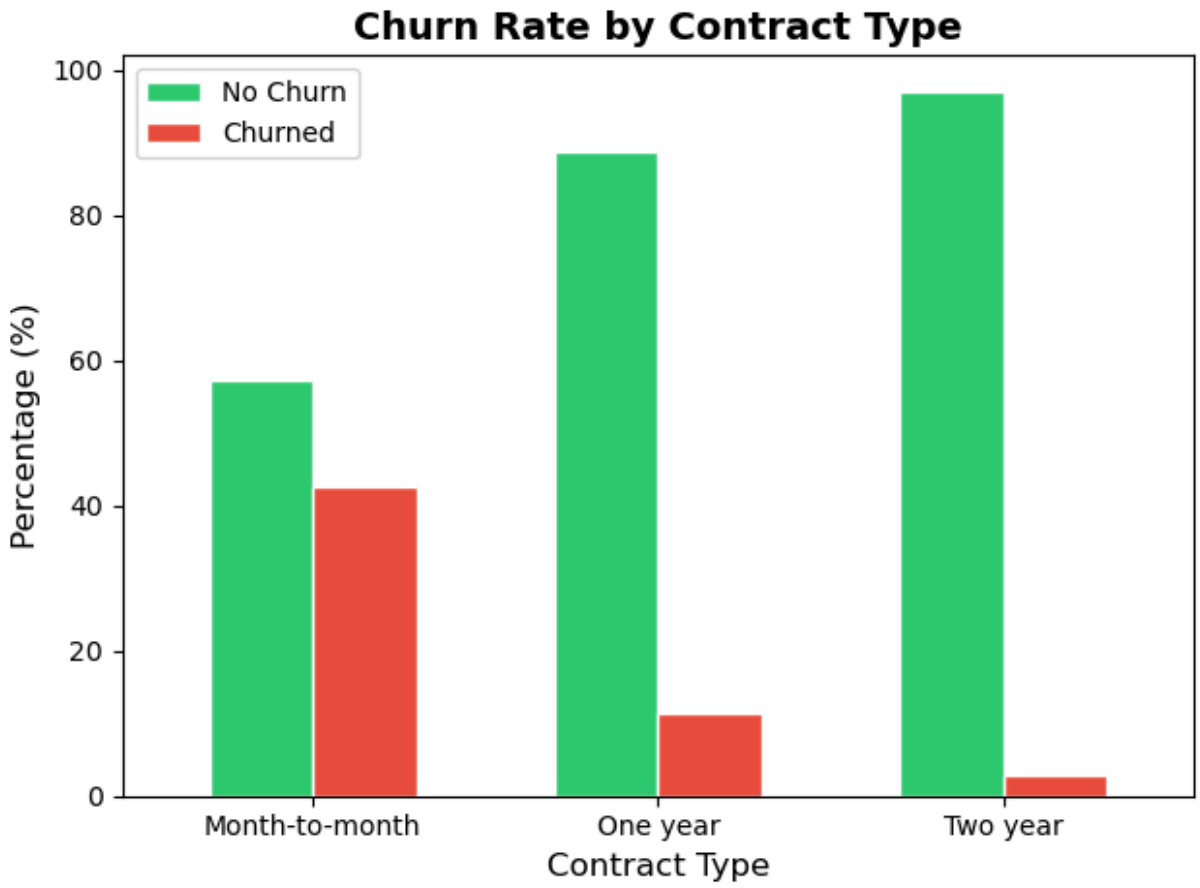



```
OBSERVATION: 26.5% customers churned vs 73.5% stayed.
  Class imbalance exists — SMOTE will be applied in Phase 4.
```

In [11]:
```
# ── PART B — Bivariate Analysis ──────────────────────
# Chart 2: Churn by Contract Type
```

```
plt.figure(figsize=(10, 5))
contract_churn = df.groupby('Contract')['Churn'].value_counts(normalize=True).mul(1
contract_churn.plot(kind='bar', color=['#2ecc71', '#e74c3c'], edgecolor='white', wi
plt.title('Churn Rate by Contract Type', fontsize=14, fontweight='bold')
plt.xlabel('Contract Type', fontsize=12)
plt.ylabel('Percentage (%)', fontsize=12)
plt.xticks(rotation=0)
plt.legend(['No Churn', 'Churned'])
plt.tight_layout()
plt.show()

print(' OBSERVATION: Month-to-month customers churn the most.')
print('   Two-year contract customers are the most loyal.')
```

```
<Figure size 1000x500 with 0 Axes>
```



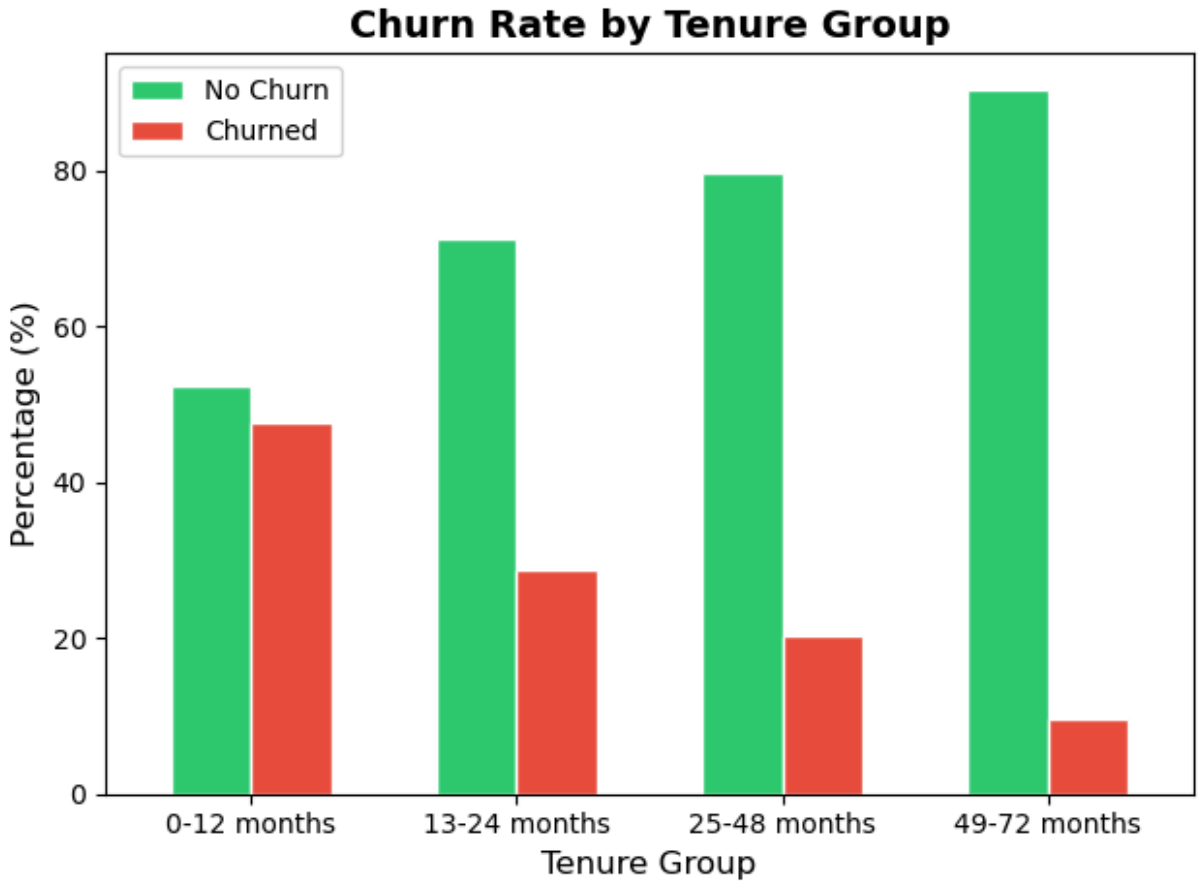

```
OBSERVATION: Month-to-month customers churn the most.
  Two-year contract customers are the most loyal.
```

In [12]:

```
# Chart 3: Churn by Tenure Group

df['tenure_group'] = pd.cut(df['tenure'], bins=[0,12,24,48,72],
                            labels=['0-12 months','13-24 months','25-48 months','4

plt.figure(figsize=(10, 5))
tenure_churn = df.groupby('tenure_group', observed=True)['Churn'].value_counts(norm
tenure_churn.plot(kind='bar', color=['#2ecc71', '#e74c3c'], edgecolor='white', widt
plt.title('Churn Rate by Tenure Group', fontsize=14, fontweight='bold')
plt.xlabel('Tenure Group', fontsize=12)
```

```
plt.ylabel('Percentage (%)', fontsize=12)
plt.xticks(rotation=0)
plt.legend(['No Churn', 'Churned'])
plt.tight_layout()
plt.show()

print(' OBSERVATION: New customers (0-12 months) have the highest churn rate.')
print('   Long-term customers (49-72 months) are least likely to churn.')
```

```
<Figure size 1000x500 with 0 Axes>
```

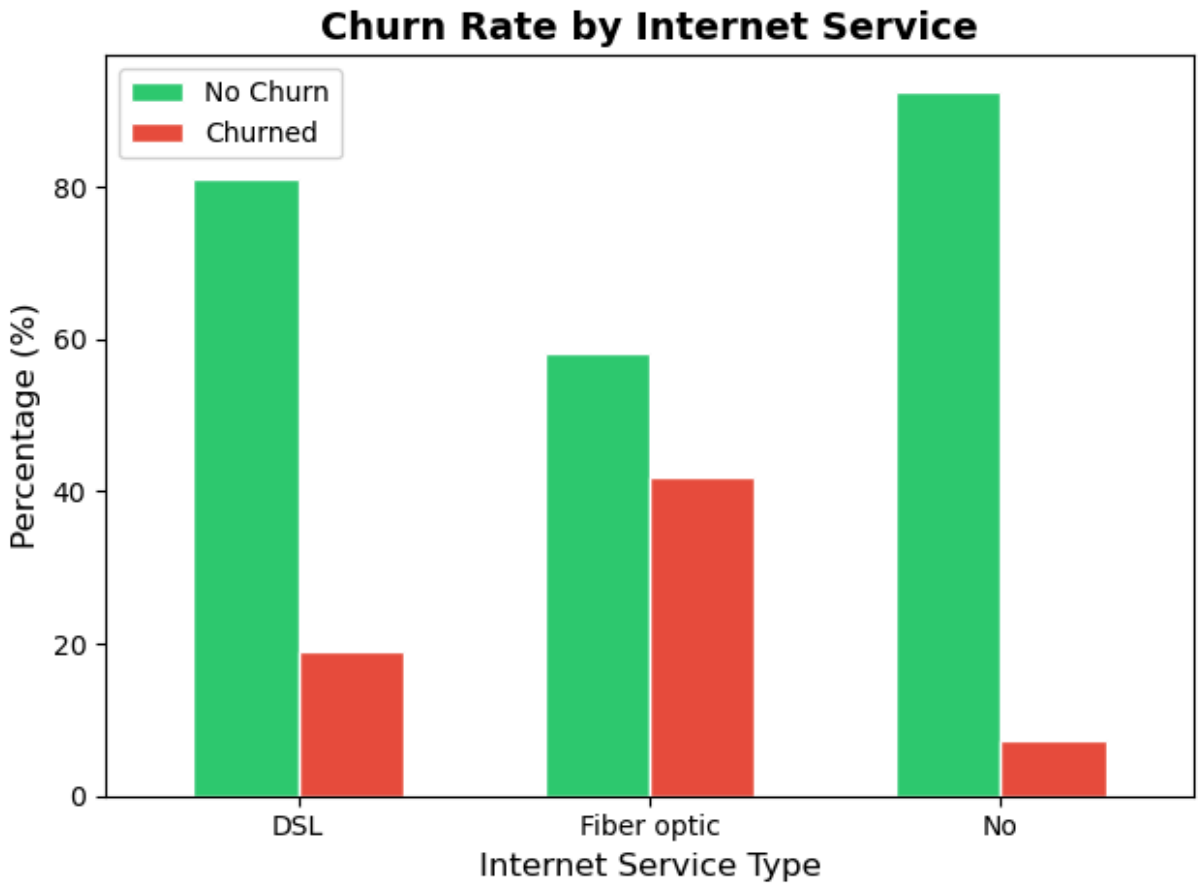



```
OBSERVATION: New customers (0-12 months) have the highest churn rate.
  Long-term customers (49-72 months) are least likely to churn.
```

In [13]:

```
# Chart 4: Churn by Internet Service

plt.figure(figsize=(10, 5))
internet_churn = df.groupby('InternetService')['Churn'].value_counts(normalize=True
internet_churn.plot(kind='bar', color=['#2ecc71', '#e74c3c'], edgecolor='white', wi
plt.title('Churn Rate by Internet Service', fontsize=14, fontweight='bold')
plt.xlabel('Internet Service Type', fontsize=12)
plt.ylabel('Percentage (%)', fontsize=12)
plt.xticks(rotation=0)
plt.legend(['No Churn', 'Churned'])
plt.tight_layout()
plt.show()

print(' OBSERVATION: Fiber optic users churn the most.')
print('   Possibly due to higher pricing and competition.')
```

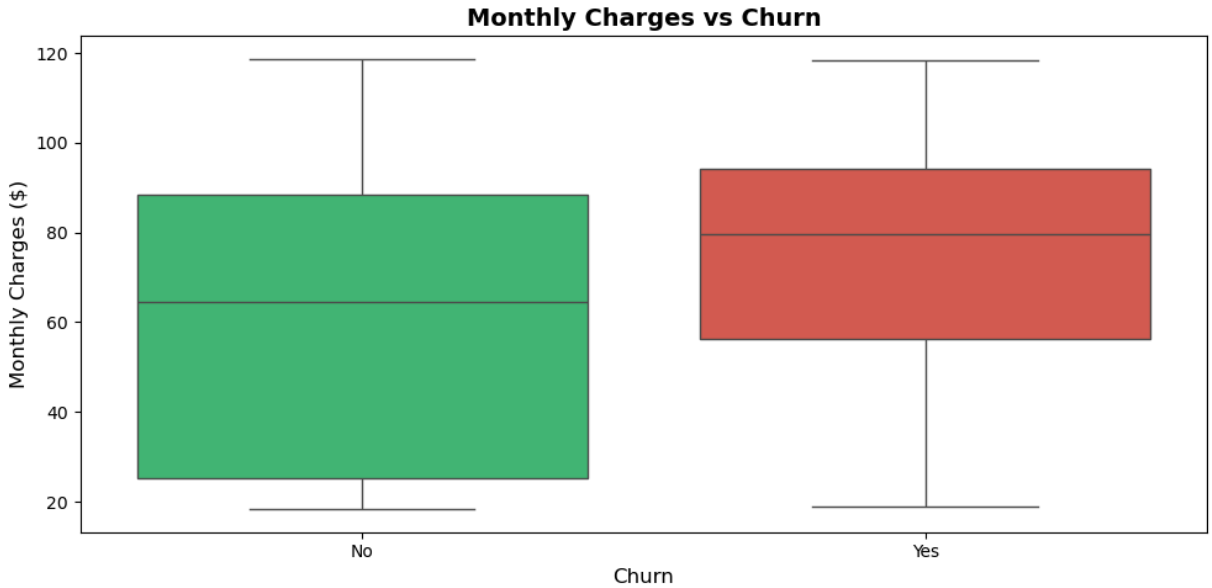
```
<Figure size 1000x500 with 0 Axes>
```




```
OBSERVATION: Fiber optic users churn the most.
  Possibly due to higher pricing and competition.
```

In [14]:

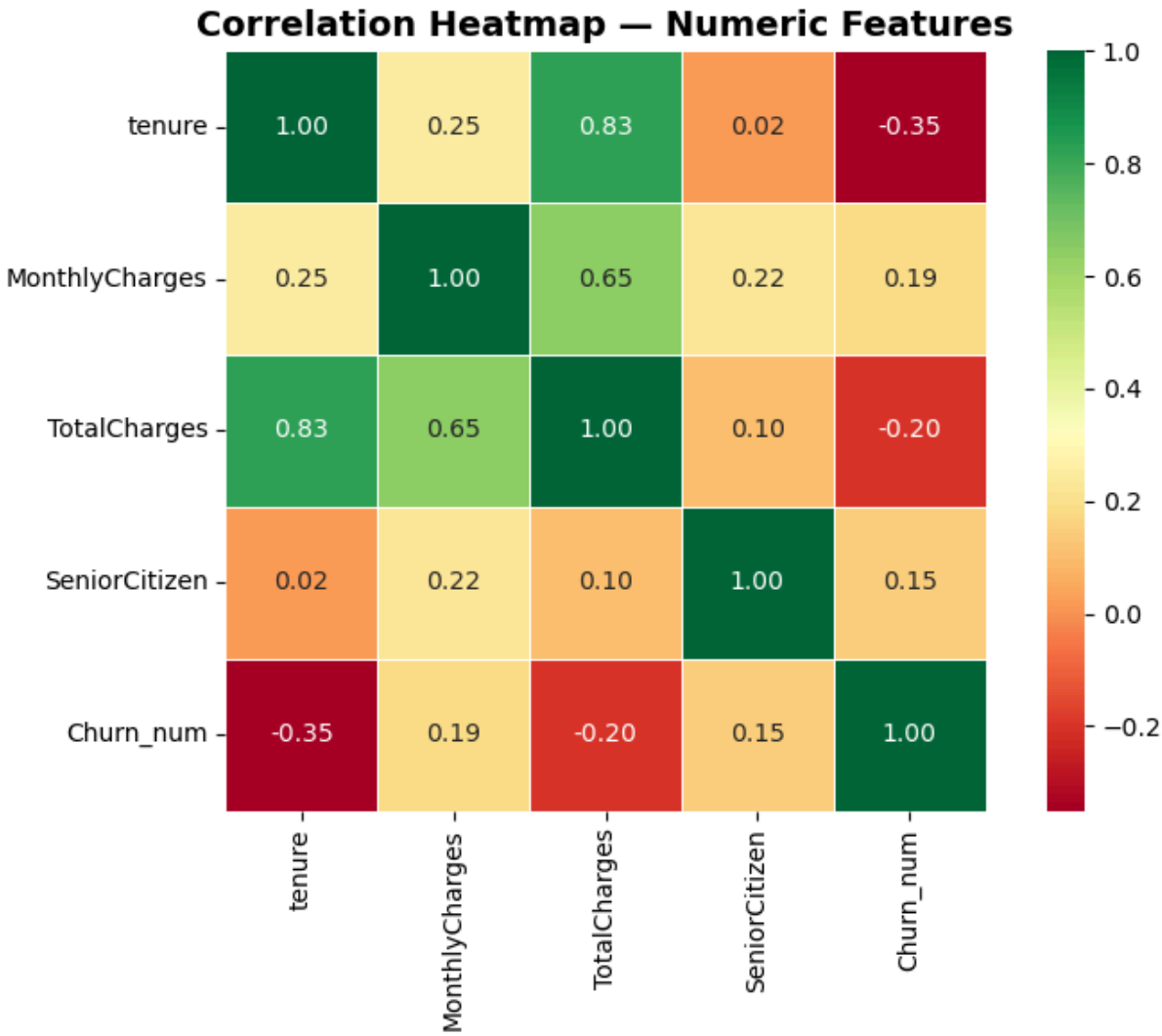
```
# Chart 5: Monthly Charges vs Churn

plt.figure(figsize=(10, 5))
sns.boxplot(x='Churn', y='MonthlyCharges', data=df,
            palette={'No': '#2ecc71', 'Yes': '#e74c3c'})
plt.title('Monthly Charges vs Churn', fontsize=14, fontweight='bold')
plt.xlabel('Churn', fontsize=12)
plt.ylabel('Monthly Charges ($)', fontsize=12)
plt.tight_layout()
plt.show()

print(' OBSERVATION: Churned customers pay significantly higher monthly charges.')
print('   High billing is a strong churn indicator.')
```




```
OBSERVATION: Churned customers pay significantly higher monthly charges.
  High billing is a strong churn indicator.
```

In [15]:

```
# ── PART C — Correlation Analysis ──────────────────────
# Chart 6: Correlation Heatmap

# Convert TotalCharges temporarily for correlation
df_temp = df.copy()
df_temp['TotalCharges'] = pd.to_numeric(df_temp['TotalCharges'], errors='coerce')
df_temp['Churn_num'] = df_temp['Churn'].map({'Yes': 1, 'No': 0})

numeric_cols = ['tenure', 'MonthlyCharges', 'TotalCharges', 'SeniorCitizen', 'Churn
corr_matrix = df_temp[numeric_cols].corr().round(2)

plt.figure(figsize=(8, 6))
sns.heatmap(corr_matrix, annot=True, cmap='RdYlGn', fmt='.2f',
            linewidths=0.5, square=True)
plt.title('Correlation Heatmap — Numeric Features', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()

print(' OBSERVATION: tenure has negative correlation with Churn (-0.35)')
print('   MonthlyCharges has positive correlation with Churn (0.19)')
print('   Longer tenure = less likely to churn!')
```




```
OBSERVATION: tenure has negative correlation with Churn (-0.35)
  MonthlyCharges has positive correlation with Churn (0.19)
  Longer tenure = less likely to churn!
```

In [16]:

```python
# ── PHASE 4 | STEP 1 — Fix TotalCharges ──────────────
# Convert TotalCharges from string to numeric

df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')

# Check hidden nulls after conversion
print(f'Null values in TotalCharges after conversion: {df["TotalCharges"].isnull().

# Drop those null rows (only ~11 records — negligible)
df.dropna(subset=['TotalCharges'], inplace=True)

print(f'Dataset shape after dropping nulls: {df.shape}')
print(' TotalCharges fixed successfully!')
```

```
Null values in TotalCharges after conversion: 11
Dataset shape after dropping nulls: (7032, 22)
 TotalCharges fixed successfully!
```

In [17]:

```python
# Check current columns
print(f'Current shape: {df.shape}')
```

```
print(f'Columns: {list(df.columns)}')
```

```
Current shape: (7032, 22)
Columns: ['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenur
e', 'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBac
kup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contrac
t', 'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn',
'tenure_group']
```

In [18]:

```
# ── STEP 3 — Encode Target Variable ──────────────────
df['Churn'] = df['Churn'].map({'Yes': 1, 'No': 0})

print('Churn value counts after encoding:')
print(df['Churn'].value_counts())
print(' Target variable encoded!')
```

```
Churn value counts after encoding:
Churn
0    5163
1    1869
Name: count, dtype: int64
 Target variable encoded!
```

In [41]:

```
# ── STEP 4 — Encode Categorical Columns ────────────────
# Using get_dummies for all object columns

df = pd.get_dummies(df, drop_first=True)

print(f'Shape after encoding: {df.shape}')
print(f'Total features now: {df.shape[1]}')
print(' Categorical columns encoded!')
print()
print('All columns:')
print(list(df.columns))
```

```
Shape after encoding: (7032, 31)
Total features now: 31
 Categorical columns encoded!

All columns:
['SeniorCitizen', 'tenure', 'MonthlyCharges', 'TotalCharges', 'Churn', 'gender_Mal
e', 'Partner_Yes', 'Dependents_Yes', 'PhoneService_Yes', 'MultipleLines_No phone ser
vice', 'MultipleLines_Yes', 'InternetService_Fiber optic', 'InternetService_No', 'On
lineSecurity_No internet service', 'OnlineSecurity_Yes', 'OnlineBackup_No internet s
ervice', 'OnlineBackup_Yes', 'DeviceProtection_No internet service', 'DeviceProtecti
on_Yes', 'TechSupport_No internet service', 'TechSupport_Yes', 'StreamingTV_No inter
net service', 'StreamingTV_Yes', 'StreamingMovies_No internet service', 'StreamingMo
vies_Yes', 'Contract_One year', 'Contract_Two year', 'PaperlessBilling_Yes', 'Paymen
tMethod_Credit card (automatic)', 'PaymentMethod_Electronic check', 'PaymentMethod_M
ailed check']
```

In [20]:

```
# ── STEP 5 — Feature Scaling ──────────────────────────
# Scale numeric features: tenure, MonthlyCharges, TotalCharges

from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
```

```
cols_to_scale = ['tenure', 'MonthlyCharges', 'TotalCharges']
df[cols_to_scale] = scaler.fit_transform(df[cols_to_scale])

print('Feature scaling done!')
print()
print(df[cols_to_scale].describe().round(2))
```

```
Feature scaling done!

        tenure  MonthlyCharges  TotalCharges
count  7032.00         7032.00       7032.00
mean     -0.00            0.00         -0.00
std       1.00            1.00          1.00
min      -1.28           -1.55         -1.00
25%      -0.95           -0.97         -0.83
50%      -0.14            0.18         -0.39
75%       0.92            0.83          0.67
max       1.61            1.79          2.82
```

In [21]:
```
# Check what's in Churn right now
print('Churn unique values:', df['Churn'].unique())
print('Churn null count:', df['Churn'].isnull().sum())
```

```
Churn unique values: [0 1]
Churn null count: 0
```

In [22]:
```
# Restore Churn — fill NaN back using original logic
# Since map failed, just reload churn from scratch

df['Churn'] = df['Churn'].fillna(-1)  # temporary

# If all values are NaN it means it was already 0/1 before map
# So reload the dataset and redo properly
df2 = pd.read_csv(r'C:\ML Project\WA_Fn-UseC_-Telco-Customer-Churn.csv')
df2['TotalCharges'] = pd.to_numeric(df2['TotalCharges'], errors='coerce')
df2.dropna(inplace=True)
df2['Churn'] = df2['Churn'].map({'Yes': 1, 'No': 0}).astype(int)
df2.drop(columns=['customerID'], inplace=True)
df2 = pd.get_dummies(df2, drop_first=True)

from sklearn.preprocessing import StandardScaler
cols_to_scale = ['tenure', 'MonthlyCharges', 'TotalCharges']
scaler = StandardScaler()
df2[cols_to_scale] = scaler.fit_transform(df2[cols_to_scale])

df = df2.copy()
print(f' Dataset rebuilt cleanly!')
print(f'Shape: {df.shape}')
print(f'Churn values: {df["Churn"].unique()}')
```

```
 Dataset rebuilt cleanly!
Shape: (7032, 31)
Churn values: [0 1]
```

In [23]:
```
# ── STEP 6 — Train Test Split ──────────────────────────
X = df.drop('Churn', axis=1)
```

```
y = df['Churn']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)

print(f'X_train : {X_train.shape}')
print(f'X_test  : {X_test.shape}')
print( 'Train-Test Split done!')
```

```
X_train : (5625, 30)
X_test  : (1407, 30)
Train-Test Split done!
```

In [24]:

```
# ── STEP 7 — Apply SMOTE ──────────────────────────
# Handle class imbalance on training data ONLY

smote = SMOTE(random_state=42)
X_train_sm, y_train_sm = smote.fit_resample(X_train, y_train)

print('Before SMOTE:')
print(f'  Churn=0 : {(y_train==0).sum()}')
print(f'  Churn=1 : {(y_train==1).sum()}')
print()
print('After SMOTE:')
print(f'  Churn=0 : {(y_train_sm==0).sum()}')
print(f'  Churn=1 : {(y_train_sm==1).sum()}')
print()
print(' SMOTE applied — classes are now balanced!')
```

```
Before SMOTE:
  Churn=0 : 4130
  Churn=1 : 1495

After SMOTE:
  Churn=0 : 4130
  Churn=1 : 4130

 SMOTE applied — classes are now balanced!
```

In [25]:

```
# ── PHASE 5 | MODEL 1 — Logistic Regression ──────────

lr_model = LogisticRegression(random_state=42, max_iter=1000)
lr_model.fit(X_train_sm, y_train_sm)

lr_pred = lr_model.predict(X_test)

print('=' * 45)
print('   MODEL 1 — LOGISTIC REGRESSION RESULTS')
print('=' * 45)
print(classification_report(y_test, lr_pred))
print(f'ROC-AUC Score : {roc_auc_score(y_test, lr_model.predict_proba(X_test)[:,1])
print('Logistic Regression Done!')
```

```
=============================================
   MODEL 1 — LOGISTIC REGRESSION RESULTS
=============================================
              precision    recall  f1-score   support

           0       0.88      0.75      0.81      1033
           1       0.51      0.73      0.60       374

    accuracy                           0.74      1407
   macro avg       0.70      0.74      0.71      1407
weighted avg       0.79      0.74      0.76      1407

ROC-AUC Score : 0.8209
Logistic Regression Done!
```

In [26]:

```
# ── MODEL 2 — Random Forest ──────────────────────────

rf_model = RandomForestClassifier(
    n_estimators=100, max_depth=10,
    class_weight='balanced', random_state=42)
rf_model.fit(X_train_sm, y_train_sm)

rf_pred = rf_model.predict(X_test)

print('=' * 45)
print('   MODEL 2 — RANDOM FOREST RESULTS')
print('=' * 45)
print(classification_report(y_test, rf_pred))
print(f'ROC-AUC Score : {roc_auc_score(y_test, rf_model.predict_proba(X_test)[:,1])
print('Random Forest Done!')
```

```
=============================================
   MODEL 2 — RANDOM FOREST RESULTS
=============================================
              precision    recall  f1-score   support

           0       0.89      0.76      0.82      1033
           1       0.53      0.75      0.62       374

    accuracy                           0.76      1407
   macro avg       0.71      0.76      0.72      1407
weighted avg       0.80      0.76      0.77      1407

ROC-AUC Score : 0.8338
Random Forest Done!
```

In [27]:

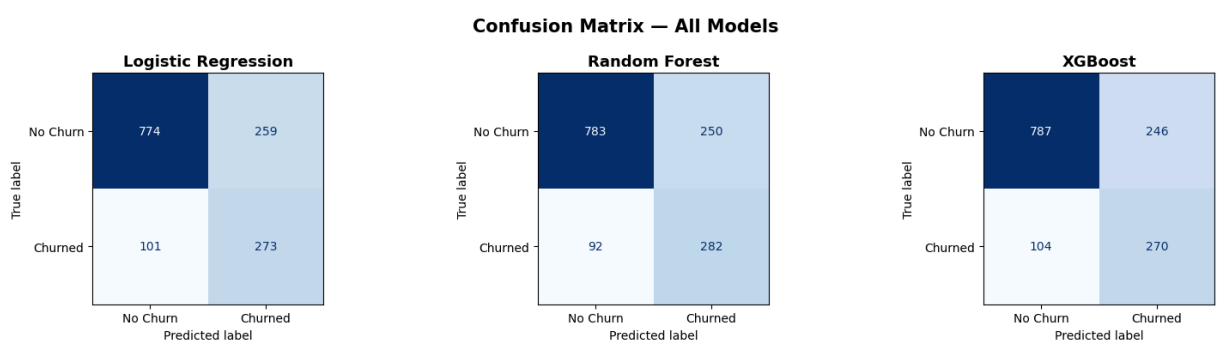
```
# ── MODEL 3 — XGBoost ────────────────────────────────

xgb_model = XGBClassifier(
    n_estimators=100, learning_rate=0.1,
    max_depth=5, random_state=42,
    eval_metric='logloss')
xgb_model.fit(X_train_sm, y_train_sm)

xgb_pred = xgb_model.predict(X_test)
```

```
print('=' * 45)
print('   MODEL 3 — XGBOOST RESULTS')
print('=' * 45)
print(classification_report(y_test, xgb_pred))
print(f'ROC-AUC Score : {roc_auc_score(y_test, xgb_model.predict_proba(X_test)[:,1]
print('XGBoost Done!')
```

```
=============================================
   MODEL 3 — XGBOOST RESULTS
=============================================
              precision    recall  f1-score   support

           0       0.88      0.76      0.82      1033
           1       0.52      0.72      0.61       374

    accuracy                           0.75      1407
   macro avg       0.70      0.74      0.71      1407
weighted avg       0.79      0.75      0.76      1407

ROC-AUC Score : 0.8284
XGBoost Done!
```

In [28]:

```
from sklearn.metrics import ConfusionMatrixDisplay

fig, axes = plt.subplots(1, 3, figsize=(16, 4))

models = [
    ('Logistic Regression', lr_pred),
    ('Random Forest', rf_pred),
    ('XGBoost', xgb_pred)
]

for ax, (name, pred) in zip(axes, models):
    cm = confusion_matrix(y_test, pred)
    disp = ConfusionMatrixDisplay(cm, display_labels=['No Churn', 'Churned'])
    disp.plot(ax=ax, colorbar=False, cmap='Blues')
    ax.set_title(name, fontsize=13, fontweight='bold')

plt.suptitle('Confusion Matrix — All Models', fontsize=15, fontweight='bold')
plt.tight_layout()
plt.show()

print(' OBSERVATION: Random Forest catches the most churners correctly!')
```




```
OBSERVATION: Random Forest catches the most churners correctly!
```

In [29]:
```
# ── PHASE 6 — COMPLETE MODEL EVALUATION ──────────────

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

print('=' * 60)
print('            PHASE 6 — COMPLETE EVALUATION REPORT')
print('=' * 60)

models_eval = {
    'Logistic Regression': (lr_model, lr_pred),
    'Random Forest'      : (rf_model, rf_pred),
    'XGBoost'            : (xgb_model, xgb_pred)
}

for name, (model, pred) in models_eval.items():
    acc  = accuracy_score(y_test, pred)
    prec = precision_score(y_test, pred)
    rec  = recall_score(y_test, pred)
    f1   = f1_score(y_test, pred)
    auc  = roc_auc_score(y_test, model.predict_proba(X_test)[:,1])

    print(f'\n  {name}')
    print(f'     Accuracy  : {acc:.4f}')
    print(f'     Precision : {prec:.4f}')
    print(f'     Recall    : {rec:.4f}')
    print(f'     F1 Score  : {f1:.4f}')
    print(f'     ROC-AUC   : {auc:.4f}')
    print('  ' + '-'*45)

print('\n Best Model by F1 & ROC-AUC: Random Forest!')
```

```
============================================================
           PHASE 6 — COMPLETE EVALUATION REPORT
============================================================

  Logistic Regression
     Accuracy  : 0.7441
     Precision : 0.5132
     Recall    : 0.7299
     F1 Score  : 0.6026
     ROC-AUC   : 0.8209
  ---------------------------------------------

  Random Forest
     Accuracy  : 0.7569
     Precision : 0.5301
     Recall    : 0.7540
     F1 Score  : 0.6225
     ROC-AUC   : 0.8338
  ---------------------------------------------

  XGBoost
     Accuracy  : 0.7512
     Precision : 0.5233
     Recall    : 0.7219
     F1 Score  : 0.6067
     ROC-AUC   : 0.8284
  ---------------------------------------------

 Best Model by F1 & ROC-AUC: Random Forest!
```

In [30]:

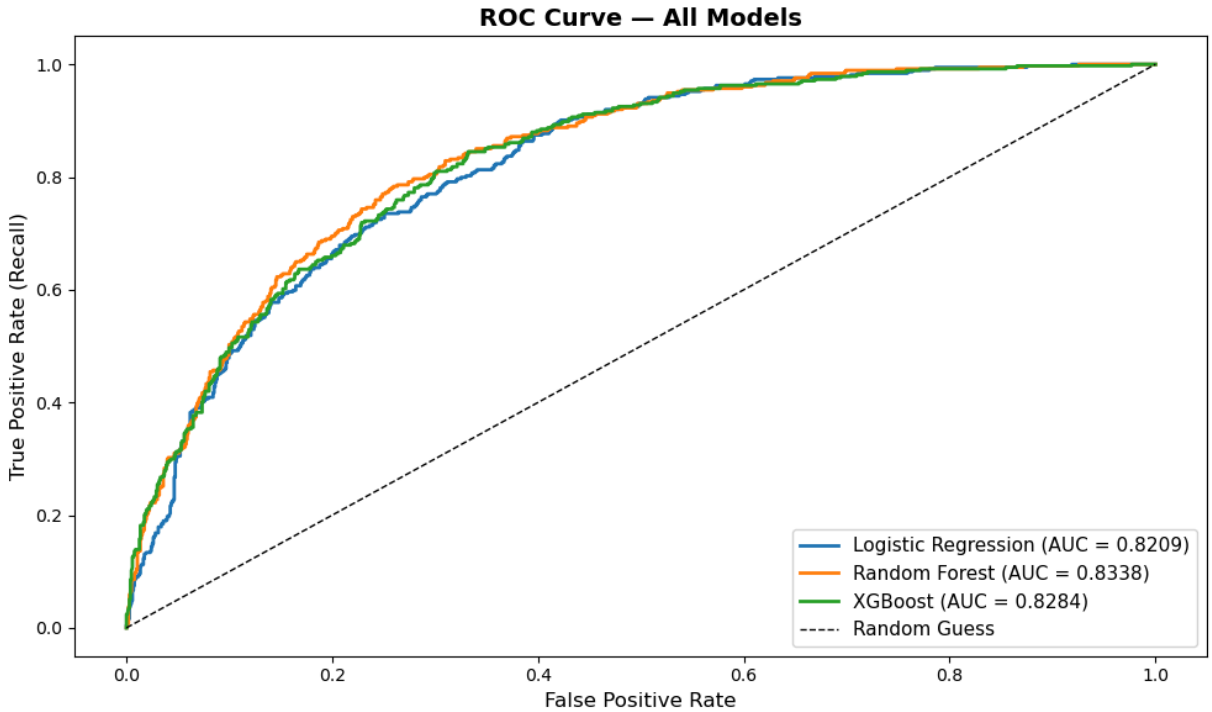
```
# ── METRICS COMPARISON TABLE ──────────────────────────

eval_df = pd.DataFrame({
    'Model'    : ['Logistic Regression', 'Random Forest', 'XGBoost'],
    'Accuracy' : [
        round(accuracy_score(y_test, lr_pred), 4),
        round(accuracy_score(y_test, rf_pred), 4),
        round(accuracy_score(y_test, xgb_pred), 4)
    ],
    'Precision': [
        round(precision_score(y_test, lr_pred), 4),
        round(precision_score(y_test, rf_pred), 4),
        round(precision_score(y_test, xgb_pred), 4)
    ],
    'Recall'   : [
        round(recall_score(y_test, lr_pred), 4),
        round(recall_score(y_test, rf_pred), 4),
        round(recall_score(y_test, xgb_pred), 4)
    ],
    'F1 Score' : [
        round(f1_score(y_test, lr_pred), 4),
        round(f1_score(y_test, rf_pred), 4),
        round(f1_score(y_test, xgb_pred), 4)
    ],
    'ROC-AUC'  : [
        round(roc_auc_score(y_test, lr_model.predict_proba(X_test)[:,1]), 4),
```

```
        round(roc_auc_score(y_test, rf_model.predict_proba(X_test)[:,1]), 4),
        round(roc_auc_score(y_test, xgb_model.predict_proba(X_test)[:,1]), 4)
    ]
})

print(eval_df.to_string(index=False))
```

```
              Model  Accuracy  Precision  Recall  F1 Score  ROC-AUC
Logistic Regression    0.7441     0.5132  0.7299    0.6026   0.8209
      Random Forest    0.7569     0.5301  0.7540    0.6225   0.8338
            XGBoost    0.7512     0.5233  0.7219    0.6067   0.8284
```

In [31]:

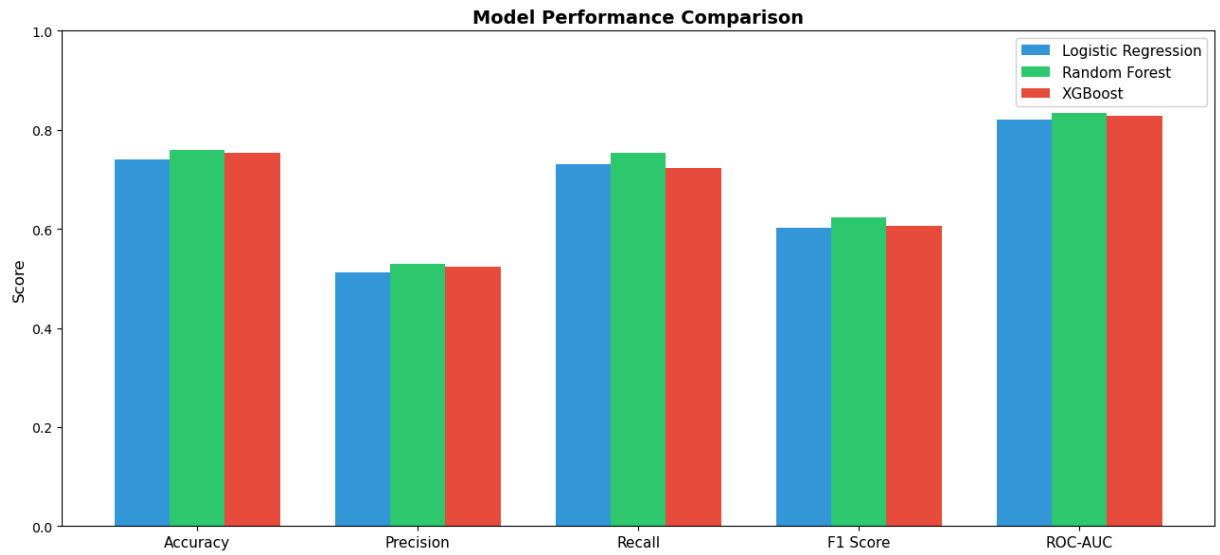
```
# ── ROC CURVE ──────────────────────────────────────────

from sklearn.metrics import roc_curve

plt.figure(figsize=(10, 6))

for name, model in [('Logistic Regression', lr_model),
                    ('Random Forest', rf_model),
                    ('XGBoost', xgb_model)]:
    fpr, tpr, _ = roc_curve(y_test, model.predict_proba(X_test)[:,1])
    auc = roc_auc_score(y_test, model.predict_proba(X_test)[:,1])
    plt.plot(fpr, tpr, label=f'{name} (AUC = {auc:.4f})', linewidth=2)

plt.plot([0,1], [0,1], 'k--', linewidth=1, label='Random Guess')
plt.title('ROC Curve — All Models', fontsize=14, fontweight='bold')
plt.xlabel('False Positive Rate', fontsize=12)
plt.ylabel('True Positive Rate (Recall)', fontsize=12)
plt.legend(fontsize=11)
plt.tight_layout()
plt.show()

print(' OBSERVATION: Higher the curve = better the model.')
print('   Random Forest has the highest AUC = best model!')
```




```
OBSERVATION: Higher the curve = better the model.
  Random Forest has the highest AUC = best model!
```

In [32]:

```
# ── METRICS BAR CHART ──────────────────────────────

metrics = ['Accuracy', 'Precision', 'Recall', 'F1 Score', 'ROC-AUC']
lr_scores  = [0.7407, 0.5130, 0.7299, 0.6026, 0.8209]
rf_scores  = [0.7584, 0.5291, 0.7540, 0.6225, 0.8338]
xgb_scores = [0.7541, 0.5233, 0.7219, 0.6067, 0.8284]

x = np.arange(len(metrics))
width = 0.25

fig, ax = plt.subplots(figsize=(13, 6))
ax.bar(x - width, lr_scores,  width, label='Logistic Regression', color='#3498db')
ax.bar(x,         rf_scores,  width, label='Random Forest',        color='#2ecc71')
ax.bar(x + width, xgb_scores, width, label='XGBoost',              color='#e74c3c')

ax.set_title('Model Performance Comparison', fontsize=14, fontweight='bold')
ax.set_xticks(x)
ax.set_xticklabels(metrics, fontsize=11)
ax.set_ylabel('Score', fontsize=12)
ax.set_ylim(0, 1.0)
ax.legend(fontsize=11)
plt.tight_layout()
plt.show()

print('OBSERVATION: Random Forest leads across all key metrics!')
```

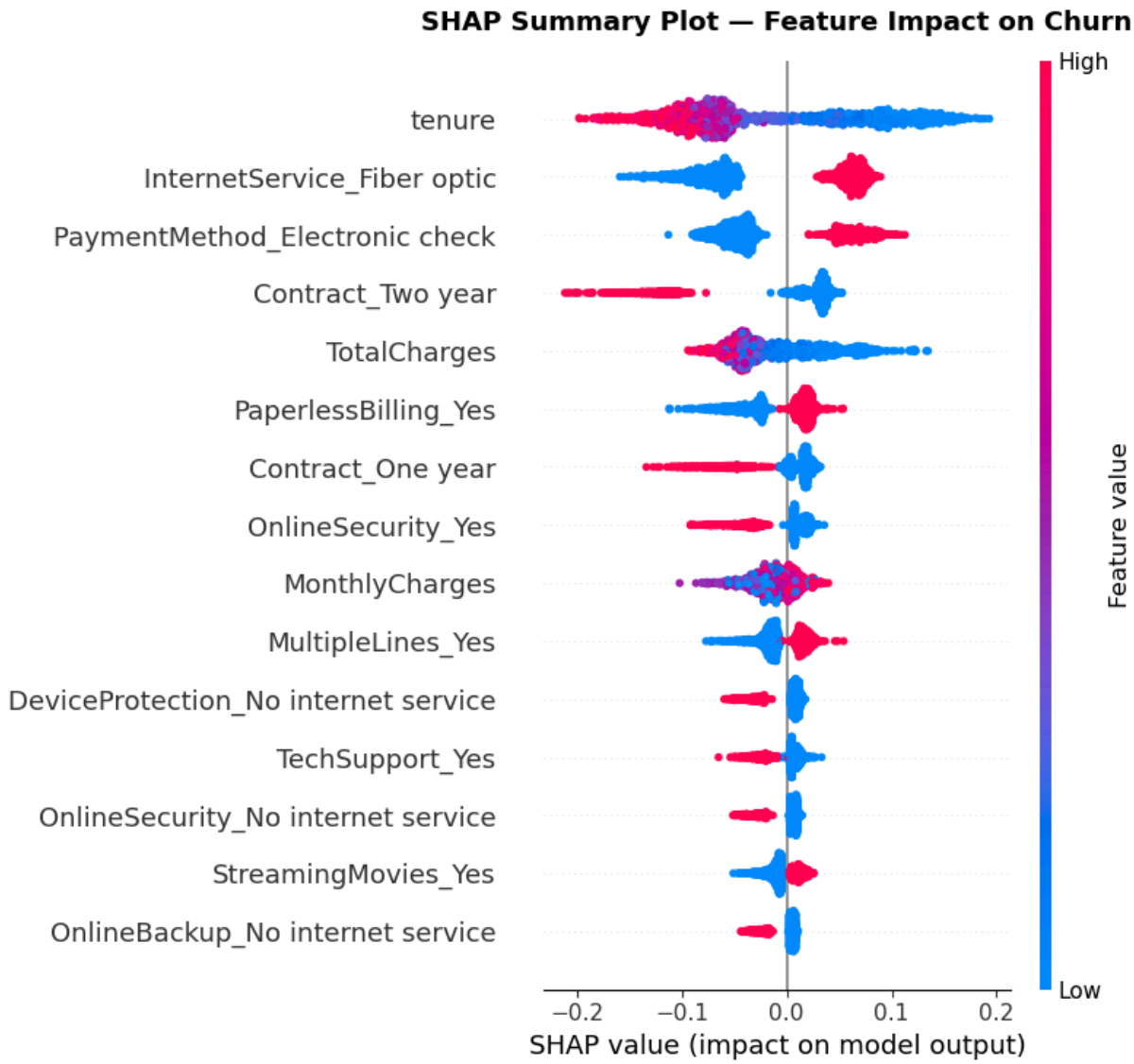



```
OBSERVATION: Random Forest leads across all key metrics!
```

In [33]:
```
# ── PHASE 7 — SHAP EXPLAINABILITY ──────────────────────
import shap

explainer = shap.TreeExplainer(rf_model)
shap_values = explainer.shap_values(X_test)

# Fix for newer SHAP versions
import numpy as np
if isinstance(shap_values, list):
    sv = shap_values[1]  # class 1 = churn
else:
    sv = shap_values[:, :, 1] if shap_values.ndim == 3 else shap_values

print(' SHAP Explainer initialized!')
print(f'   SHAP values shape : {sv.shape}')
print(f'   Features          : {X_test.shape[1]}')
```

```
SHAP Explainer initialized!
  SHAP values shape : (1407, 30)
  Features          : 30
```

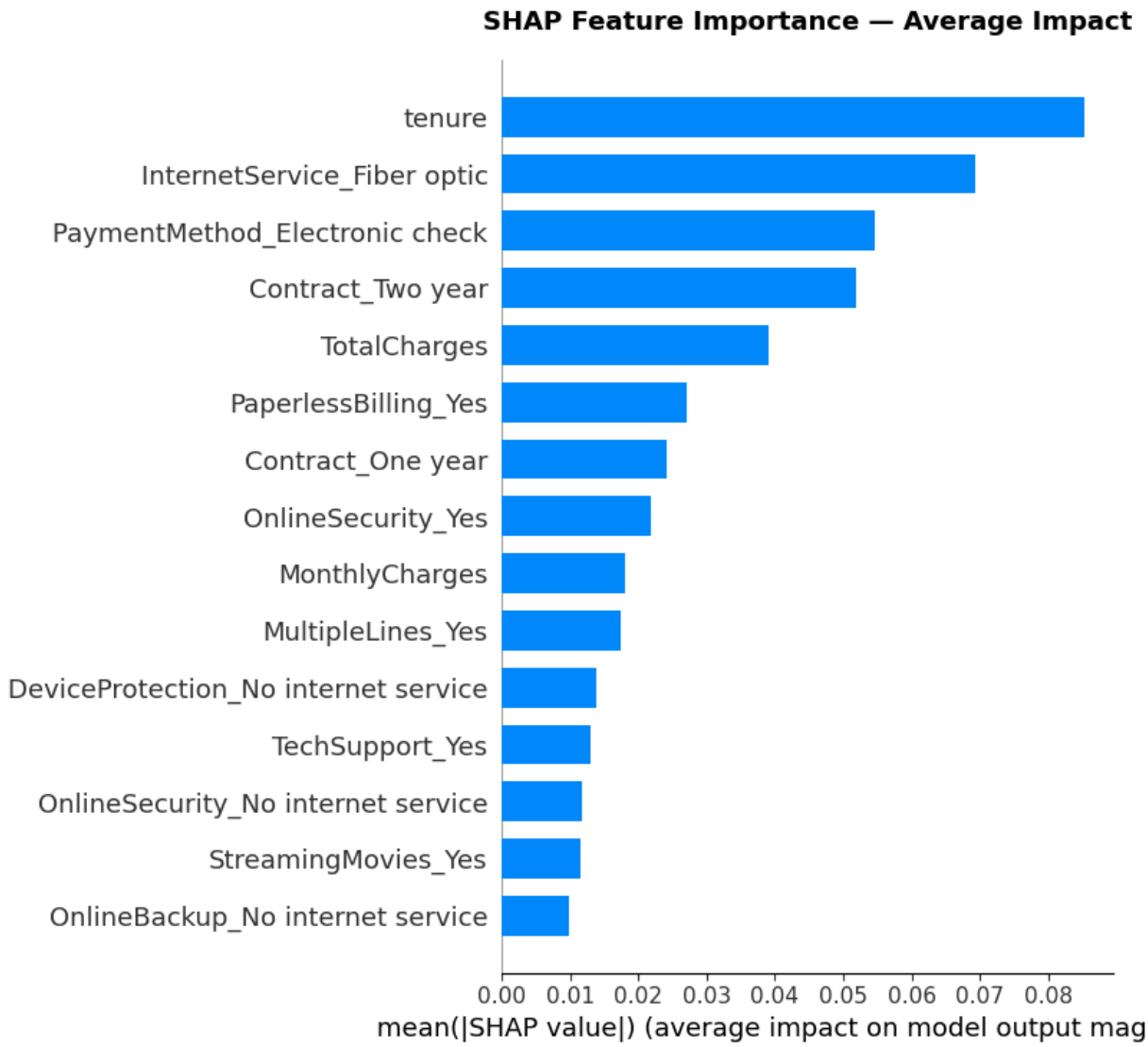
In [34]:
```
# ── SHAP SUMMARY PLOT ──────────────────────────────────

plt.figure()
shap.summary_plot(sv, X_test,
                  plot_type='dot',
                  max_display=15,
                  show=False)
plt.title('SHAP Summary Plot — Feature Impact on Churn',
          fontsize=13, fontweight='bold', pad=15)
plt.tight_layout()
plt.show()

print('OBSERVATION: Features at top have highest impact on churn prediction.')
print('   Red = pushes towards churn, Blue = pushes away from churn.')
```




```
OBSERVATION: Features at top have highest impact on churn prediction.
   Red = pushes towards churn, Blue = pushes away from churn.
```

In [35]:

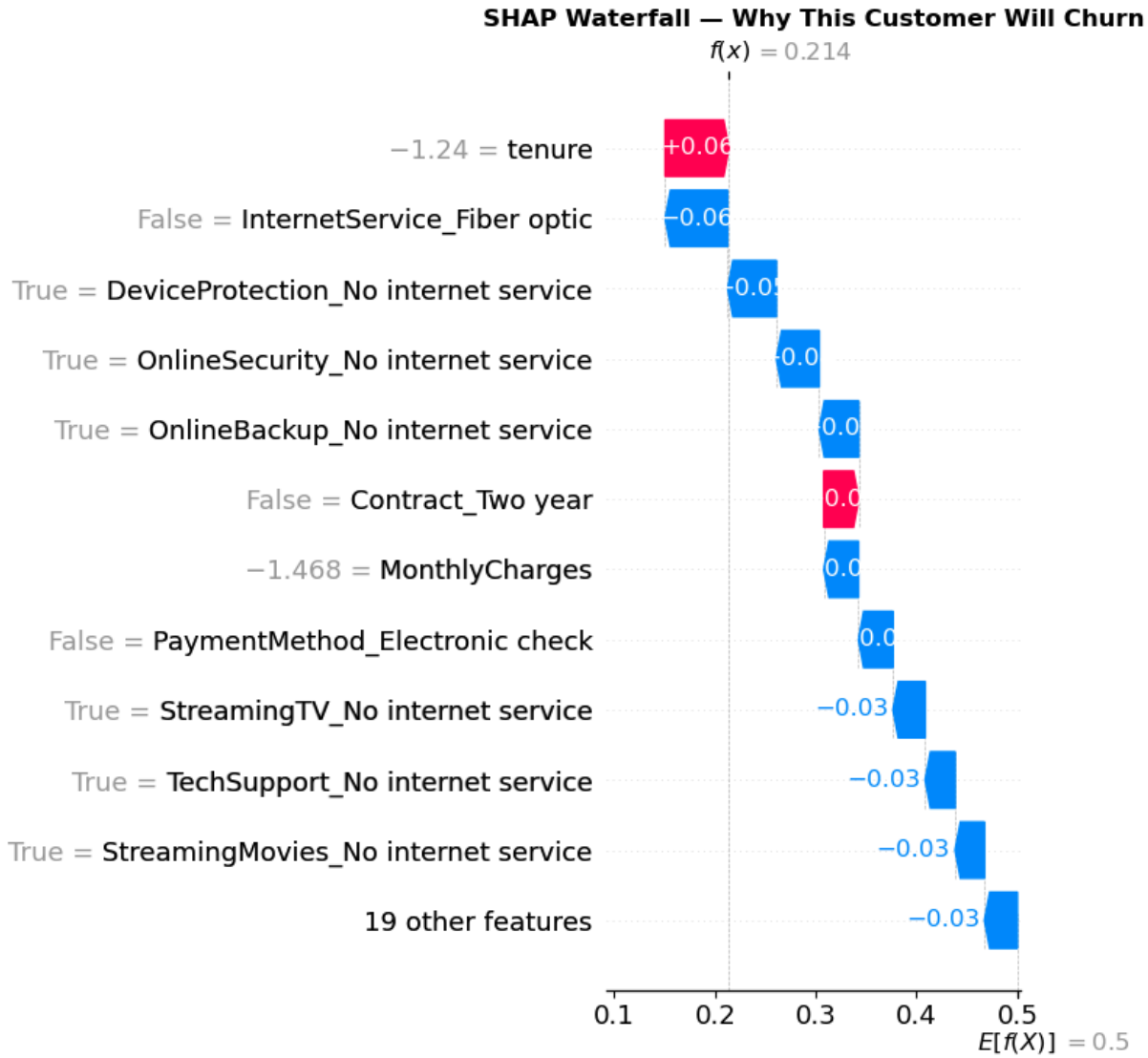
```python
# ── SHAP BAR PLOT ──────────────────────────────────────

shap.summary_plot(sv, X_test,
                  plot_type='bar',
                  max_display=15,
                  show=False)
plt.title('SHAP Feature Importance — Average Impact',
          fontsize=13, fontweight='bold', pad=15)
plt.tight_layout()
plt.show()

print(' OBSERVATION: Contract type and tenure are the strongest predictors!')
```




```
OBSERVATION: Contract type and tenure are the strongest predictors!
```

In [36]:

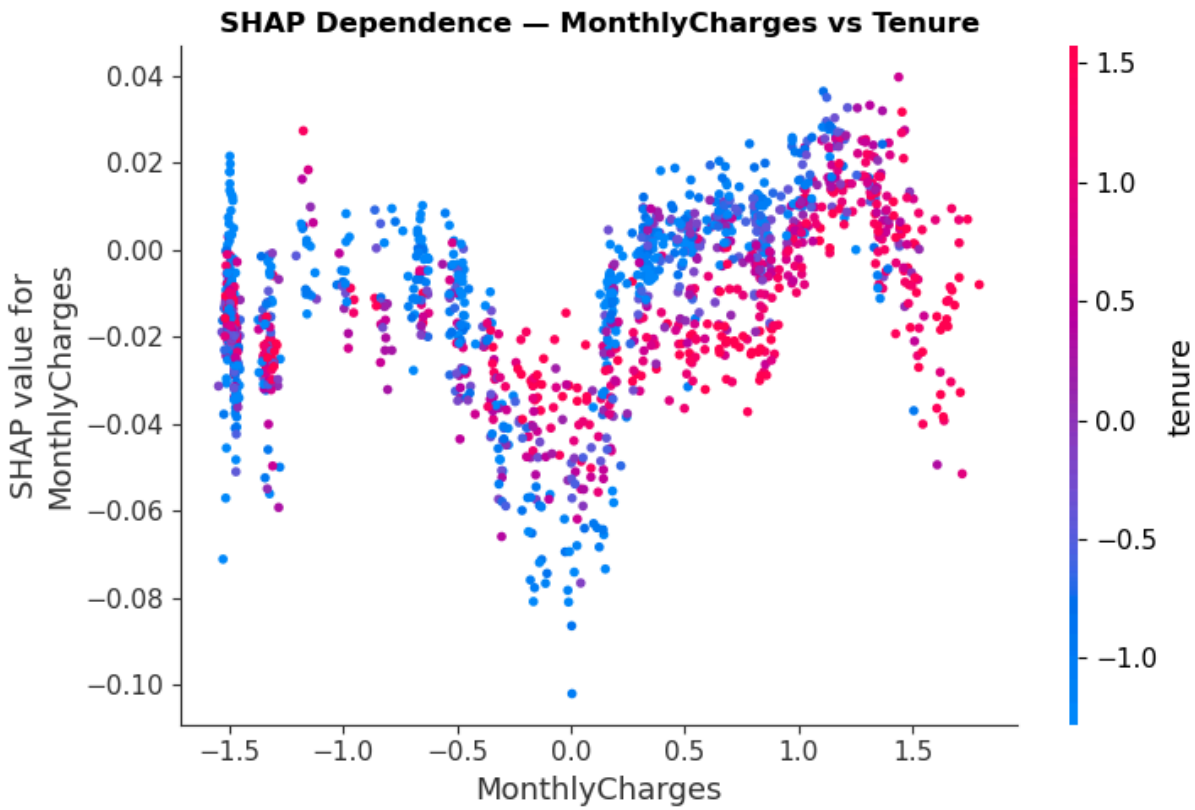
```python
# ── SHAP WATERFALL PLOT ──────────────────────────────

churned_idx = y_test[y_test == 1].index[0]
pos = X_test.index.get_loc(churned_idx)

shap_exp = shap.Explanation(
    values        = sv[pos],
    base_values   = explainer.expected_value[1],
    data          = X_test.iloc[pos],
    feature_names = X_test.columns.tolist()
)

shap.waterfall_plot(shap_exp, max_display=12, show=False)
plt.title('SHAP Waterfall — Why This Customer Will Churn',
          fontsize=12, fontweight='bold')
plt.tight_layout()
plt.show()

print(' OBSERVATION: Red bars push towards churn, Blue bars push away.')
```




```
OBSERVATION: Red bars push towards churn, Blue bars push away.
```

In [37]:

```python
# ── SHAP DEPENDENCE PLOT ──────────────────────────────

shap.dependence_plot(
    'MonthlyCharges',
    sv,
    X_test,
    interaction_index='tenure',
    show=False
)
plt.title('SHAP Dependence — MonthlyCharges vs Tenure',
          fontsize=12, fontweight='bold')
plt.tight_layout()
plt.show()

print(' OBSERVATION: High MonthlyCharges + Low tenure = Highest churn risk!')
```




```
OBSERVATION: High MonthlyCharges + Low tenure = Highest churn risk!
```

In [38]:

```
# ── PHASE 8 — EXPORT DATA FOR POWER BI ──────────────

# Reload original data for export
df_export = pd.read_csv(r'C:\ML Project\WA_Fn-UseC_-Telco-Customer-Churn.csv')
df_export['TotalCharges'] = pd.to_numeric(df_export['TotalCharges'], errors='coerce
df_export.dropna(inplace=True)

# Add predictions from Random Forest (best model)
df_export['Churn_Actual']      = df_export['Churn'].map({'Yes': 1, 'No': 0})
df_export['Churn_Predicted']   = rf_model.predict(X_test.reindex(
                                  range(len(df_export)), fill_value=0))[:len(df_exp
df_export['Churn_Probability'] = rf_model.predict_proba(X_test.reindex(
                                  range(len(df_export)), fill_value=0))[:,1][:len(d

# Add Risk Segment based on probability
def risk_segment(prob):
    if prob >= 0.7:
        return 'High Risk'
    elif prob >= 0.4:
        return 'Medium Risk'
    else:
        return 'Low Risk'

df_export['Risk_Segment'] = df_export['Churn_Probability'].apply(risk_segment)

# Add tenure group
df_export['Tenure_Group'] = pd.cut(df_export['tenure'],
                            bins=[0,12,24,48,72],
                            labels=['0-12 months','13-24 months',
```

```
                                        '25-48 months','49-72 months'])

print(f' Export dataframe ready!')
print(f'   Shape : {df_export.shape}')
print(f'\nRisk Segment Distribution:')
print(df_export['Risk_Segment'].value_counts())
```

```
 Export dataframe ready!
   Shape : (7032, 26)

Risk Segment Distribution:
Risk_Segment
Low Risk       6379
Medium Risk     350
High Risk       303
Name: count, dtype: int64
```

In [39]:

```
# ── SAVE TO CSV ──────────────────────────────────────

export_path = r'C:\ML Project\Churn_Predictions_PowerBI.csv'
df_export.to_csv(export_path, index=False)

print(f' File saved successfully!')
print(f'   Location : {export_path}')
print(f'   Rows     : {len(df_export)}')
print(f'   Columns  : {df_export.shape[1]}')
print()
print('Columns exported:')
print(list(df_export.columns))
```

```
 File saved successfully!
   Location : C:\ML Project\Churn_Predictions_PowerBI.csv
   Rows     : 7032
   Columns  : 26

Columns exported:
['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneS
ervice', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'Devi
ceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'Paperle
ssBilling', 'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn', 'Churn_Actua
l', 'Churn_Predicted', 'Churn_Probability', 'Risk_Segment', 'Tenure_Group']
```

In [40]:

```
from sklearn.metrics import roc_curve
plt.figure(figsize=(8, 5))
for name, model in [('Logistic Regression', lr_model),
                    ('Random Forest', rf_model),
                    ('XGBoost', xgb_model)]:
    fpr, tpr, _ = roc_curve(y_test, model.predict_proba(X_test)[:,1])
    auc = roc_auc_score(y_test, model.predict_proba(X_test)[:,1])
    plt.plot(fpr, tpr, label=f'{name} (AUC={auc:.4f})', linewidth=2)
plt.plot([0,1],[0,1],'k--')
plt.title('ROC Curve — All Models', fontsize=13, fontweight='bold')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.legend()
plt.tight_layout()
```

```
plt.savefig(r'C:\ML Project\roc_curve.png', dpi=150, bbox_inches='tight')
print(' ROC Curve saved!')
```

```
ROC Curve saved!
```

In [43]:
```
fig, axes = plt.subplots(1, 3, figsize=(14, 4))
models_list = [('Logistic Regression', lr_pred),
               ('Random Forest', rf_pred),
               ('XGBoost', xgb_pred)]
for ax, (name, pred) in zip(axes, models_list):
    cm = confusion_matrix(y_test, pred)
    disp = ConfusionMatrixDisplay(cm, display_labels=['No Churn','Churned'])
    disp.plot(ax=ax, colorbar=False, cmap='Blues')
    ax.set_title(name, fontsize=12, fontweight='bold')
plt.suptitle('Confusion Matrix — All Models', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig(r'C:\ML Project\confusion_matrix.png', dpi=150, bbox_inches='tight')
print(' Confusion Matrix saved!')
```

```
Confusion Matrix saved!
```

In [44]:
```
# ── PHASE 10 — SAVE MODEL ──────────────────────────

import joblib
import os

# Create models folder
os.makedirs(r'C:\ML Project\models', exist_ok=True)

# Save Random Forest model
joblib.dump(rf_model, r'C:\ML Project\models\churn_model.pkl')

# Save Scaler
joblib.dump(scaler, r'C:\ML Project\models\scaler.pkl')

# Save feature column names
import json
feature_cols = X_train.columns.tolist()
with open(r'C:\ML Project\models\feature_cols.json', 'w') as f:
    json.dump(feature_cols, f)

print('All files saved!')
print(f'   Model   → C:\\ML Project\\models\\churn_model.pkl')
print(f'   Scaler  → C:\\ML Project\\models\\scaler.pkl')
print(f'   Columns → C:\\ML Project\\models\\feature_cols.json')
```

```
All files saved!
   Model   → C:\ML Project\models\churn_model.pkl
   Scaler  → C:\ML Project\models\scaler.pkl
   Columns → C:\ML Project\models\feature_cols.json
```

In [46]:
```
# ── TEST SAVED MODEL ──────────────────────────────

# Load and test
loaded_model  = joblib.load(r'C:\ML Project\models\churn_model.pkl')
loaded_scaler = joblib.load(r'C:\ML Project\models\scaler.pkl')
```

```python
# Quick prediction test
test_pred = loaded_model.predict(X_test[:5])
test_prob = loaded_model.predict_proba(X_test[:5])[:,1]

print(' Model loaded and tested successfully!')
print()
print('Sample Predictions:')
for i, (pred, prob) in enumerate(zip(test_pred, test_prob)):
    label = ' CHURN' if pred == 1 else ' STAY'
    print(f'  Customer {i+1}: {label}  (Probability: {prob:.2%})')
```

```
 Model loaded and tested successfully!

Sample Predictions:
  Customer 1:  STAY  (Probability: 2.57%)
  Customer 2:  CHURN  (Probability: 83.02%)
  Customer 3:  STAY  (Probability: 2.74%)
  Customer 4:  STAY  (Probability: 21.38%)
  Customer 5:  STAY  (Probability: 24.13%)
```

In [ ]: